

Miruoto : Sports event atmosphere visual rendering through real-time image and sound processing system

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Figure 1: Miruoto : a real time system to render sounds and ambiance on sports events videos through onomatopoeia

ABSTRACT

Did you already imagine how would it be to watch a sport match without sounds?

You would miss all this specific sport related sounds but also mostly miss a big part of the atmosphere present in the stadium, that is particular to live events. This is what happens to most Deaf and Hard of Hearing persons. Towards Tokyo 2025 Deaflympics, we developed an AI-based system able to recognize sounds and players motion to render in real time sound related Onomatopoeia over the match video as one could see in Comics or Manga.

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1 INTRODUCTION

Every hearing person would tell you the importance of sounds during sport events. Each sports have its own characteristic sound : Tennis racket and ball hitting sound, Baseball ball hitting sound,

Volleyball smash sound, Table Tennis ball sound, Bowling strike sound, Basketball dribble sound, etc. On the top of these sounds during a live event match the ambiance sounds such as applause, cheering, booing, whistling, etc.. Toward the Tokyo 2025 Deaflympics, an international multi-sport event where all players have hearing impairment related disabilities, we started wondering how would a Deaf and Hard of Hearing public reacts to watching live sports. Here we present a system that mimics sound in real time through rendering Onomatopoeia overlay on the sport match video. To achieve this we used AI-based sound and vision recognition system.

1.1 Onomatopoeia

Onomatopoeia designate a word that mimics a sound, they are mostly used in spoken languages English/European languages or to write sounds with words in comics [Ye 2023] In American and Europeans Comics onomatopoeia are sound based and are used to describe a variety of sound like "Bang","Boom","Splash" etc... this is referred as Sound Symbolism [Gerd 2014; Svantesson 2017]

In the Japanese language Onomatopoeia can describe sounds, but also describe emotions, feelings or physics related sensation like softness, roughness or temperature.[Akita 2009; Hamano 2000]

Added to the fact that Japan has about 60 percent of its 20-49 years old population reading Manga [Statista 2022] makes this population used to frequently read or see in a written form Onomatopoeia, even Hearing impaired persons. It had also been shown that Hearing impaired person would rely less in onomatopoeia in language [Manar 2021] but with some learning they grasped it [Namatame 2019]. A few research have been done in order to superimpose automatically Onomatopoeia on Computer Graphics

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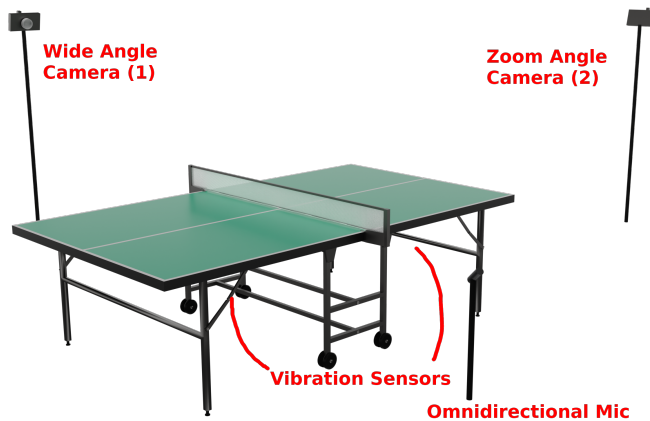


Figure 2: Camera, Vibration sensors and Mic setup.

[Fukusato and Morishima 2014], or changing drawing characteristics through Onomatopoeia [Kambara and Tsukada 2010].

1.2 Miruoto System Description

Our Miruoto system (<https://miruoto.jp/>) consists in overlaying sounds-related onomatopoeia on real-time sport video. For this paper we will focus on the Table Tennis sport, and the sounds related to this sport. The main features are :

- **A real-time AI recognition** of sound and pictures.
- **An intuitive rendering adapted to Deaf, Hard of Hearing, and Hearing persons** through Onomatopoeia designed by hearing impaired designers.
- **Displaying Onomatopoeia** when the ball touches the table or the racket and when the public applause and reacts or the players shoes makes some noise
- **The Onomatopoeia position and graphics**, like font and color, vary depending from where the sound come from.
- **The Onomatopoeia size and animation** vary depending the loudness of the sound.
- **Multiple camera point of view** and adapted positioning.
- **A graphic interface** to display scores, players name, etc.

1.3 Technical aspect

The system is composed of 3 different type of sensors:

1.3.1 Cameras : The system used two cameras as described in Fig 2. The first one is a 120° camera is placed next to the Tennis Table net at a distance of 3m from the table. The view is a complete view of the table and about 2m on each side. This picture is used as a complete view of the match, and is close to the side view usually seen in TV broadcast. This image is then processed by YOLOv8 an AI network trained to find racket and players in pictures. The second one is a camera with an embedded optical zoom (5-50mm f1:1.6). It is placed to see the back of one of the player, around from 3m to 10m behind and provide the same point of view as someone sitting in the stadium watching the match.

1.3.2 Microphones : An omnidirectional microphone (*Primo EMU4785 Unit EM272*) is used to record sounds of applause, racket and ball

hit sounds and shoes squeaking sounds. It is placed next to the first camera. Sound recognition is done through a MobileNet[G. 2017] called YAMNET pretrained on AudioSet [F. 2017] and finetuned with 100ms length audio file of racket-ball impacts and squeak sounds.

1.3.3 Vibration Sensors : Two vibration sensors (*Primo MX5617*) are positioned under the table on each player side. The amplitude of the vibration emitted by the ball bounce is used to find which side the ball is currently hitting the table.

2 POSSIBLE APPLICATION

We plan to use our system for Tokyo 2025 Deaflympics in order to "show" sound and ambiance during the match that Deaf and Hard or Hearing cannot or cannot well perceive under normal condition. It could also be used during TV shows, for giving more attraction to the public to some less broadcast sports, or to teach kids some sports and make it more amusing. We already have started working on other sports for our system.

3 CONCLUSION

While searching to add some accessibility to sports to Deaf and Hard of Hearing, we developed a system that enhance user experience of watching live sport matches and that is intuitive to everyone, disregarding their hearing abilities.

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